Prediction of Electric Vehicle Range Using Machine Learning Regression Models

Pankaj
Dept. of Computer Science & Engineering
Lovely Professional University, Phagwara, India
Reg. No.: 12419598, Roll No.: 67
Supervised by: Mannet Kaur

***Abstract*—Electric vehicles (EVs) are becoming an important part of modern transportation due to their potential to reduce dependence on conventional fuels and support sustainable mobility. Electric range is an important factor in evaluating an electric vehicle because it indicates the distance a vehicle can travel using its available battery energy. This paper presents a machine learning-based approach for predicting the electric range of electric vehicles using vehicle-related attributes. Five regression algorithms are implemented and compared: Linear Regression, Decision Tree Regressor, Random Forest Regressor, K-Nearest Neighbors (KNN) Regressor, and Multi-Layer Perceptron (MLP) Neural Network. The dataset is preprocessed by removing duplicate records, handling missing values, encoding categorical attributes, and selecting relevant features. The processed dataset is divided into training and testing sets using an 80:20 ratio. Model performance is evaluated using the coefficient of determination (R^2), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE). The experimental results are used to compare the predictive performance of the selected algorithms and identify the best-performing model. This study provides a comparative framework for applying machine learning regression techniques to electric vehicle range prediction.**

***Keywords*— Electric Vehicles, Machine Learning, Electric Range Prediction, Regression Models, Random Forest, Decision Tree, K-Nearest Neighbors, Neural Network.**

I. Introduction

Electric vehicles have gained considerable attention as an alternative to conventional internal combustion engine vehicles. They use electric motors for propulsion and rely on electrical energy stored in batteries. With the increasing adoption of electric vehicles, understanding their driving range has become an important consideration for consumers, manufacturers, and researchers.

Electric range refers to the distance an electric vehicle can travel before its available battery energy becomes insufficient for continued driving. The range of a vehicle may vary according to its design, model year, vehicle category, battery technology, and other technical characteristics. Accurate range prediction can help users compare vehicles and understand differences in their expected driving capabilities.

Traditional methods of estimating vehicle range may depend on standardized testing procedures and manufacturer-provided specifications. Machine learning techniques offer an alternative approach by learning relationships between vehicle attributes and their recorded electric ranges.

This study develops and compares five machine learning regression models for predicting electric vehicle range. The selected algorithms are Linear Regression, Decision Tree Regressor, Random Forest Regressor, K-Nearest Neighbors Regressor, and Multi-Layer Perceptron Neural Network.

The study uses vehicle-related attributes, including model year, manufacturer, model, electric vehicle type, and Clean Alternative Fuel Vehicle (CAFV) eligibility. These attributes are processed and supplied to the regression algorithms to predict the target variable, Electric Range.

A. Problem Statement

Predicting electric vehicle range from vehicle-related attributes is a regression problem in which the objective is to estimate a continuous numerical value. Different machine learning algorithms may produce different prediction errors depending on the patterns present in the dataset.

Therefore, a comparative evaluation is required to understand how different regression techniques perform on the selected dataset.

B. Scope of the Study

This study focuses on predicting the recorded electric range of vehicles using selected vehicle-related attributes. It compares five regression algorithms under the same training and testing split. The analysis also includes visualizations of actual and predicted values, model errors, training loss, electric range distribution, model year, and vehicle type.

The study does not directly predict real-time driving range under changing road, weather, battery-health, or driving conditions because these variables are not included in the selected feature set.

II. Related Work

Machine learning has been applied to various prediction problems in transportation, including vehicle performance estimation, energy consumption analysis, and driving-range estimation. Regression algorithms are particularly useful when the output variable is continuous and numerical.

Linear Regression is a fundamental statistical learning technique that estimates relationships between input variables and a target variable. It provides a simple baseline for evaluating more complex models.

Decision Tree Regression predicts numerical outcomes by dividing the feature space into smaller regions using decision rules. It can represent nonlinear relationships but may overfit the training data if its complexity is not controlled.

Random Forest Regression combines multiple decision trees to generate predictions. By aggregating predictions from several trees, it can capture nonlinear relationships and reduce dependence on a single tree.

K-Nearest Neighbors Regression estimates a target value using observations that are close to the input sample according to a distance measure. Its performance depends on the distribution and scaling of the features.

Multi-Layer Perceptron Neural Networks use interconnected computational units and nonlinear activation functions to learn complex patterns in data. Their performance may depend on the network architecture, feature representation, and training parameters.

In this study, these five algorithms are evaluated using the same dataset preparation process and train-test split. This provides a common basis for comparing their performance on electric vehicle range prediction.

III. Dataset And Preprocessing

A. Dataset Description

The study uses a CSV dataset containing electric vehicle information. The dataset is loaded into a Pandas Data Frame for preprocessing and analysis.

The dataset contains vehicle-related attributes, including model year, manufacturer, model, electric vehicle type, and CAFV eligibility. The target variable is Electric Range, which the models are trained to predict.

The dataset is loaded from a file named export.csv. Its original source, collection date, geographical coverage, and total number of records should be documented after verification.

B. Feature Selection

Five input features are selected from the dataset to predict the electric range of electric vehicles.
Model Year: Represents the year associated with the vehicle model. For example, 2018, 2023, and 2025.
Make: Represents the vehicle manufacturer, such as Tesla, Ford, and Mazda.
Model: Represents the specific vehicle model, such as Model 3, Mustang Mach-E, and CX-90.
Electric Vehicle Type: Specifies the category of the electric vehicle. The dataset includes Battery Electric Vehicles (BEVs) and Plug-in Hybrid Electric Vehicles (PHEVs).
Clean Alternative Fuel Vehicle (CAFV) Eligibility: Indicates the vehicle's eligibility classification, such as Clean Alternative Fuel Vehicle Eligible, Eligibility Unknown, or Not Eligible Due to Low Battery Range.
The target variable is Electric Range, which represents the electric range value recorded for each vehicle in the dataset. The machine learning models use the five selected input features to predict this target variable.

C. Data Cleaning

Duplicate records are removed to reduce repeated observations in the dataset. The Model Year and Electric Range columns are converted to numeric data types. Values less than or equal to zero in these two columns are treated as missing values.

Rows with missing Electric Range values are removed because the target value is required during supervised model training. Missing values in Model Year are replaced with the median of the available values.

For the remaining categorical features, missing values are replaced with the label "Unknown".

D. Categorical Encoding

Machine learning algorithms require numerical representations of the input features. Therefore, Label Encoding is applied to the categorical features: Make, Model, Electric Vehicle Type, and CAFV Eligibility.

Each distinct category is assigned an integer value. This allows the selected algorithms to process categorical information numerically.

However, label encoding can introduce artificial numerical ordering into categorical variables. This limitation should be considered when interpreting the model results.

E. Dataset Sampling

After preprocessing and encoding, a random sample of up to 20,000 records is selected using a fixed random state of 42. This provides a reproducible sampling procedure when the same dataset and software environment are used.

The actual number of records used for modeling depends on the number of valid observations available after cleaning.

F. Training and Testing Split

The processed dataset is divided into training and testing subsets using an 80:20 ratio. The train_test_split function is configured with a random state of 42.

The training subset is used to fit the regression models, while the testing subset is used to evaluate predictions on observations not used during model fitting.

IV. Proposed Methodology

A. System Overview

The proposed system follows a machine learning regression workflow. First, the electric vehicle dataset is loaded and cleaned. Next, the selected features are encoded and the dataset is sampled. The processed data is then divided into training and testing subsets.

Five regression models are trained on the training subset. Each model generates predictions for the testing subset. These predictions are compared with the actual Electric Range values using three evaluation metrics: R^2 Score, RMSE, and MAE.

Finally, the evaluation results are arranged in ascending order of RMSE to identify the model with the lowest observed RMSE.

B. Linear Regression

Linear Regression is used as a baseline regression algorithm. It models the relationship between input features and the target variable by estimating coefficients that minimize the residual sum of squares.

The general form of a linear regression model is:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n$$

In the implementation, the Linear Regression class from scikit-learn is trained using the training data. Predictions are generated for the testing data and evaluated using the selected metrics.

C. Decision Tree Regressor

Decision Tree Regression predicts numerical values by recursively dividing observations into groups according to feature-based decision rules. Each terminal leaf produces a

numerical prediction based on the training observations assigned to that leaf.

The implementation uses a maximum tree depth of 15 and a random state of 42. Limiting the maximum depth helps control tree complexity.

The trained model predicts electric range for the testing dataset. A comparison plot is generated to visualize actual and predicted values for the first 100 test observations, or fewer if fewer observations are available.

D. Random Forest Regressor

Random Forest Regression is an ensemble learning technique that combines predictions from multiple decision trees. The final prediction is obtained by aggregating the outputs of the individual trees.

The implementation uses 100 estimators, a maximum depth of 15, a random state of 42, and parallel processing through n_jobs=-1.

The Random Forest model is trained using the selected features and evaluated on the testing subset. Its actual-versus-predicted scatter plot helps visualize how closely the predictions follow the observed electric range values.

E. K-Nearest Neighbors Regressor

K-Nearest Neighbors Regression estimates the target value using the nearest observations in the feature space. For regression, the standard implementation predicts the average target value of the selected neighbors.

The model in this study uses five neighbors. It is trained on the training subset and then used to predict the electric range of the test observations.

A residual plot is created to examine the differences between actual and predicted values. The residual is calculated as:

$$e_i = y_i - \hat{y}_i$$

F. Multi-Layer Perceptron Neural Network

A Multi-Layer Perceptron is a feedforward neural network that learns relationships between input features and numerical outputs through training.

The implementation uses two hidden layers containing 50 and 25 neurons, respectively. The maximum number of training iterations is set to 300, and the random state is set to 42.

The model's loss curve is plotted to visualize the training loss across iterations. This curve provides information about the optimization process but does not, by itself, establish how well the model generalizes to unseen observations.

G. Model Comparison

All five models are evaluated using the same testing subset. Their R^2 Score, RMSE, and MAE values are stored in a comparison table.

The table is sorted by RMSE in ascending order. Consequently, the model with the smallest RMSE appears first in the sorted results.

The model ranked first according to RMSE is identified as the best-performing model under this particular evaluation criterion. Its R^2 Score and MAE are also reported to provide a broader assessment of predictive performance.

Regression Model	**R^2 Score**	**RMSE**	**MAE**
Random Forest	0.9968	5.44	1.06
Decision Tree	0.9967	5.51	1.06
KNN	0.9955	6.44	1.16
Linear Regression	0.7680	46.4	32.68
MLP Neural Network	0.7623	46.97	28.25

V. Performance Evalution Mertics

The performance of the machine learning regression models is evaluated using three metrics: Coefficient of Determination (R^2 Score), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE). These metrics are used to compare the predicted electric range values with the actual values and assess the accuracy of the models.

A. Coefficient of Determination (R^2 Score)

The R^2 Score measures how well a regression model explains the variation in the target variable. A higher R^2 Score generally indicates better predictive performance.

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

B. Root Mean Squared Error (RMSE))

RMSE measures the square root of the average squared difference between actual and predicted values. A lower RMSE indicates smaller prediction errors.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

C. Mean Absolute Error (MAE)

MAE measures the average absolute difference between actual and predicted values. A lower MAE indicates better prediction accuracy.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

VI. Graphs Description

This section presents the experimental results obtained from the five machine learning regression models used for electric vehicle range prediction. The performance of the models is analyzed using actual versus predicted value plots, residual analysis, training loss, and R^2 Score comparison. The results are also examined through exploratory data analysis to understand the distribution of electric range, its relationship

with model year, and the average electric range across different vehicle types. These visualizations help evaluate model performance and identify important patterns in the dataset.

Graph 1 – Linear Regression:

The Linear Regression model predicts electric range using the selected vehicle features. Figure 1 compares the actual electric range values with the predicted values. The red dashed line represents ideal predictions, where actual and predicted values are equal. The graph shows that several points deviate from the ideal line, particularly for vehicles with higher actual electric range values. This indicates that Linear Regression does not capture all the variations in the dataset effectively.

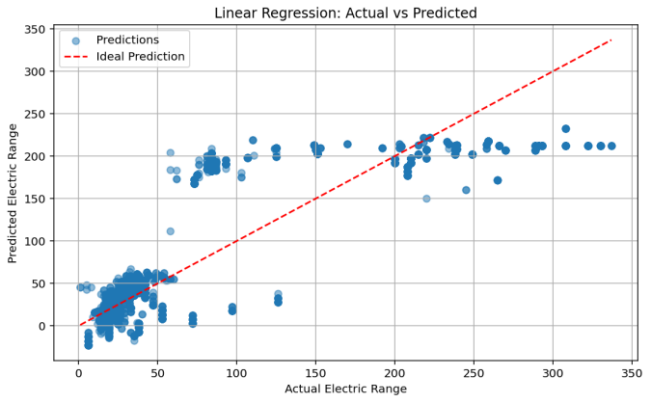


Figure 1. Linear Regression: Actual vs. Predicted Electric Range

Graph 2 – Decision Tree Regressor:

The Decision Tree Regressor is evaluated by comparing actual and predicted electric range values for the first 100 test samples. Figure 2 shows that the predicted values closely follow the actual values for most observations. In several cases, the two lines overlap, indicating small prediction errors. However, some differences remain between actual and predicted values, showing that the model does not produce perfect predictions for every sample.

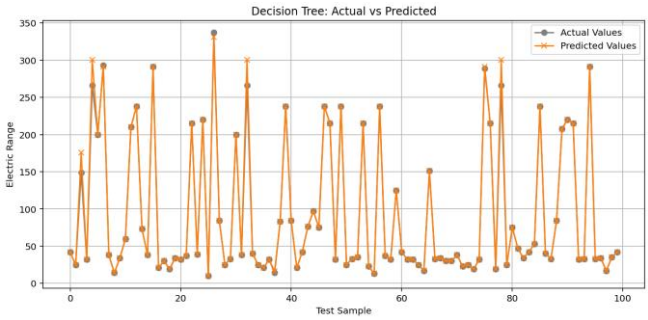


Figure 2. Decision Tree: Actual vs. Predicted Electric Range

Graph 3 – Random Forest Regressor:

Figure 3 presents the actual and predicted electric range values generated by the Random Forest Regressor. Most data points lie close to the red dashed ideal prediction line, indicating strong agreement between actual and predicted values. A few points deviate from the line, representing larger prediction errors for certain observations. The graph suggests that Random Forest performs well on the evaluated test data.

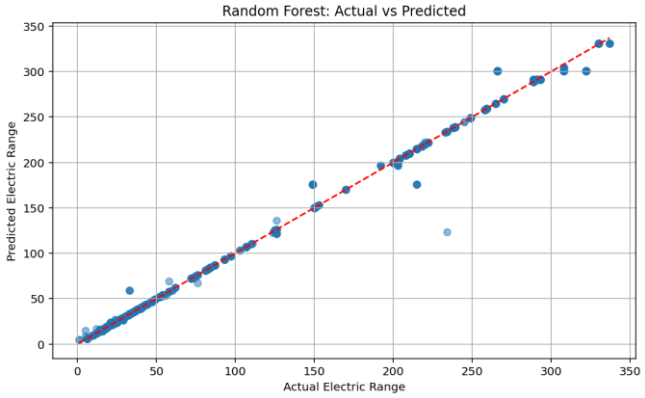


Figure 3. Random Forest: Actual vs. Predicted Electric Range

Graph 4 – K-Nearest Neighbors (KNN) Regressor:

The KNN model is analyzed using a residual plot, which displays the differences between actual and predicted electric range values. Figure 4 shows that many residuals lie close to the zero line, indicating relatively small errors for several observations. However, some points have considerably larger positive or negative residuals. These variations indicate that the model produces larger prediction errors for certain vehicles.

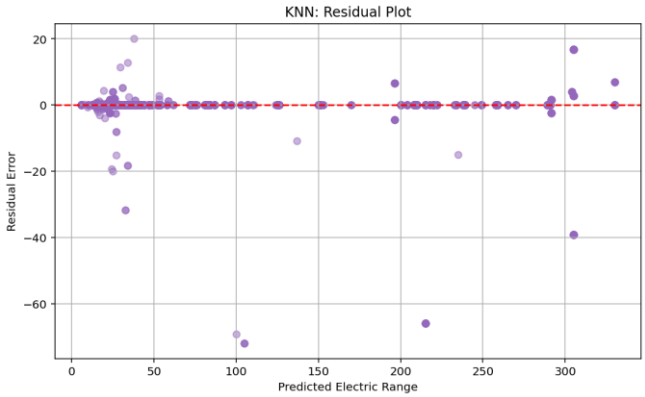


Figure 4. KNN Residual Plot

Graph 5 –Multi-Layer Perceptron (MLP) Neural Network :

Figure 5 illustrates the training loss of the MLP Neural Network across training iterations. The loss decreases significantly during the initial iterations and then gradually becomes stable. This indicates that the model reduces its training error as the optimization process progresses. However, the training loss alone cannot determine the model's performance on unseen data; therefore, test-set evaluation metrics are also required.

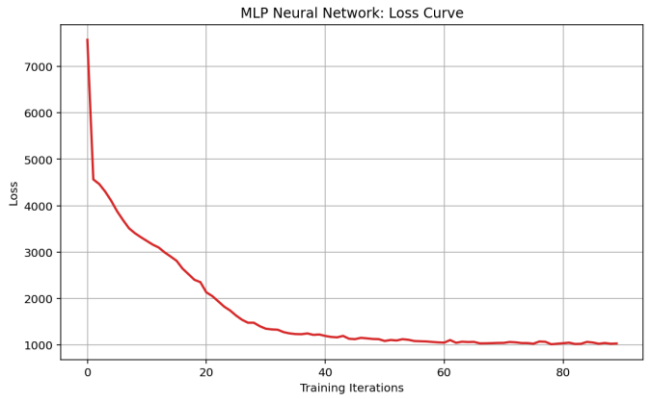


Figure 5. MLP Neural Network: Training Loss Curve

Graph 6 – R^2 Score Comparison of All Models:

Figure 6 compares the R^2 Scores of the five regression models. Random Forest achieves the highest score of 0.9968, followed by Decision Tree at 0.9967 and KNN at 0.9955. Linear Regression and MLP Neural Network achieve scores of 0.7680 and 0.7623, respectively. These results indicate that Random Forest has the highest R^2 Score among the evaluated models, although its advantage over Decision Tree and KNN is small. RMSE and MAE should also be considered when selecting the final model.

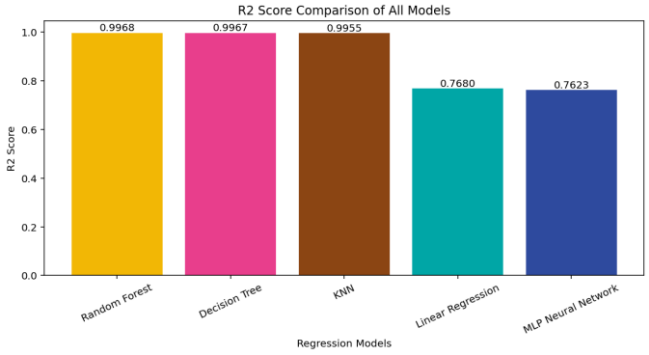


Figure 6. R^2 Score Comparison of All Regression Models

Graph 6 – Distribution of Electric Range:

Figure 7 illustrates the distribution of electric range values in the dataset using a histogram. The graph shows a high concentration of vehicles with lower electric range values, particularly between approximately 20 and 50. Another concentration appears around 200 to 250, while fewer observations occur in several intermediate ranges. This indicates that electric range values are not uniformly distributed and may reflect differences between vehicle categories and models.

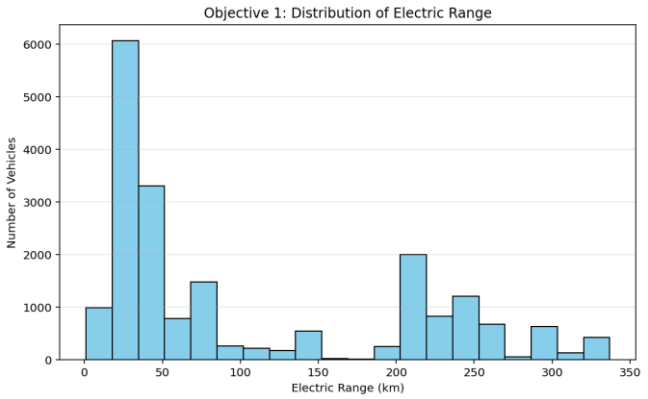


Figure 7. Distribution of Electric Range

Graph 6 – Model Year vs. Electric Range:

Figure 8 shows the relationship between vehicle model year and electric range. The graph indicates considerable variation in electric range across different model years. Some newer vehicles have higher recorded electric ranges, while many vehicles from recent years still have relatively low values. This suggests that model year alone does not fully explain electric range, and other factors, such as manufacturer, vehicle model, and electric vehicle type, may also influence the observed differences.

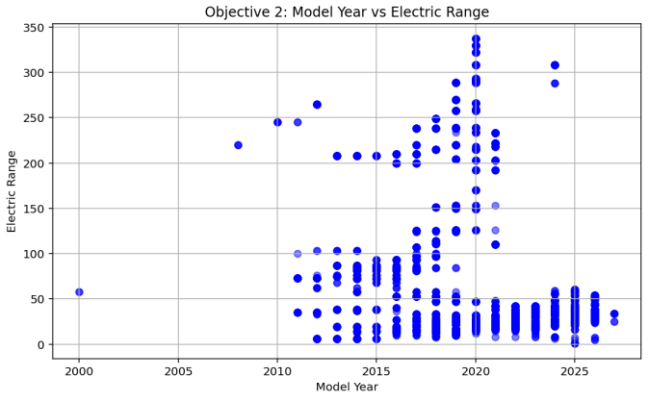


Figure 8. Model Year vs. Electric Range

Graph 6 – Average Electric Range by Vehicle Type:

Figure 9 compares the average electric range of Battery Electric Vehicles (BEVs) and Plug-in Hybrid Electric Vehicles (PHEVs). The graph shows an average value of 201.23 for BEVs and 32.13 for PHEVs. Therefore, BEVs have a substantially higher average recorded electric range than PHEVs in the processed dataset. This comparison highlights the difference between the two vehicle categories. The values should be interpreted according to the unit used in the original dataset.

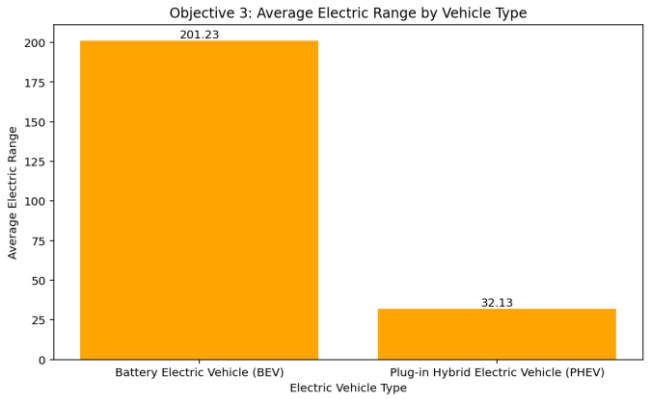


Figure 9. Average Electric Range by Vehicle Type

VII. System Overview

The proposed system aims to predict the electric range of electric vehicles using machine learning regression algorithms. The system takes vehicle-related information as input and predicts the corresponding electric range. The input features include Model Year, Make, Model, Electric Vehicle Type, and Clean Alternative Fuel Vehicle (CAFV) Eligibility.

The system begins by loading the electric vehicle dataset from a CSV file using the Pandas library. The dataset is then preprocessed by removing duplicate records, converting numerical columns into appropriate data types, handling missing values, and encoding categorical features into numerical representations. After preprocessing, a random sample of up to 20,000 records is selected for model development.

The processed dataset is divided into training and testing subsets using an 80:20 ratio. Five regression algorithms, namely Linear Regression, Decision Tree Regressor, Random Forest Regressor, K-Nearest Neighbors (KNN) Regressor, and Multi-Layer Perceptron (MLP) Neural Network, are trained using the training data. Each model generates predictions on the testing data.

The predicted values are evaluated against the actual electric range values using R^2 Score, Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE). Finally, the results are compared using performance metrics and graphical visualizations to identify the best-performing model. The system provides a comparative analysis of different machine learning approaches for electric vehicle range prediction.

VIII. Acknowledgment

I would like to express my sincere gratitude to my project guide and faculty members of the Department of Computer Science and Engineering for their valuable guidance and support throughout this project. I am also thankful to my college for providing the necessary resources and learning environment to complete this work. Finally, I would like to thank the developers of Python and its libraries, including Pandas, NumPy, Matplotlib, and Scikit-learn, which were used to implement and evaluate the machine learning models in this project.

References

The Electric Vehicle Range Prediction system uses data science and machine learning concepts to analyze electric vehicle characteristics and predict their electric range. The methodology is based on machine learning techniques and regression models discussed in scholarly research [1], [2]. The dataset used in this project contains information about electric vehicles, including model year, manufacturer, vehicle model, electric vehicle type, CAFV eligibility, and electric range [3]. Exploratory data analysis and visualization techniques are used to understand the distribution of electric range and identify patterns in the dataset [4], [5]. Linear Regression, Decision Tree, Random Forest, K-Nearest Neighbors (KNN), and Multi-Layer Perceptron (MLP) models are implemented using machine learning concepts described in [6], [7]. Model performance is evaluated using R^2 Score, Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE) to compare prediction accuracy [8]. Python libraries, including Pandas, NumPy, Matplotlib, and Scikit-learn, are used for data preprocessing, visualization, model training, and evaluation [9], [10].

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